Engagement Journal: Project SafeCart

Before I looked. Before loading the data I thought it would be so meaningful that I would not have to do much hustle. It came from an authentic source so I expected a well structured table where every column meant what its name said. That was my first prediction. My second was that chemical alerts would be a small and steady slice of our five categories across the years. My third was that China would sit far ahead of everyone else as the country of origin. I wrote all three down and committed them before running the first cell so the commit time is my evidence.

What I did. I uploaded the Excel file in Colab and read every column as text so nothing got changed quietly, which gave 31152 rows and 30 columns. I looked at the first rows and tried to say what one row is. Then I turned the two date columns into real dates and counted alerts per year. My first dead end was the alert number. I treated it as a unique key until a duplicate check showed 12 numbers used twice with different edit times. After that I counted missing values, then noticed that empty cells were only half the story because words like unknown and no brand are missing in disguise. Then I picked the five categories, built the chemical flag from the risk type text and started drawing charts. The brand histogram was unreadable at first so I moved it to a log scale. The basic coding was done by us. I used AI tools mostly for the coding and since the last two labs were on EDA they had already taught me cleaning and sorting so I knew what I was asking for. For the complex tasks and the cleaning parts we took help from AI and for the visualisation section we used it to make the charts look more appealing. We did not rely on it alone because every suggestion was read run on the raw data and sometimes thrown away when it did not fit.

What surprised me. The first surprise was the alert description. It is empty for almost every row in 2015 to 2017 and then full from 2018 onward and the barcode behaves the same way until 2019. I checked by grouping the empty cells by year and the switch was so sharp that it cannot be random. It looks like a form change not careless authorities. The second surprise was the chemical share inside our five categories. It sat around 40 percent for years then jumped to about 68 percent in 2023. I did not believe it, so I held the category mix fixed at its 2015 to 2022 shape and recomputed. The line stayed between 40 and 44 percent. The reason is that cosmetics went from roughly 5 to 11 percent of our alerts before 2022 to about half of them. It is not only mix though, because cosmetics moved from the 80s to 98 percent chemical and electricals from 2 to about 15 percent, so tagging habits probably changed as well.

What I revised. My first note in the notebook said a consumer from the United Kingdom reported the alert. The data showed that alert submitted by is a national authority and type of public says who the product is meant for, which is consumer in 30320 rows and professional in only 820. I rewrote the definition of a row. I also thought alert number was unique but 12 repeats were older versions of the same alert so I kept the latest edit and went from 31152 to 31140 rows. For cleaning of data I first counted only blank cells as missing but placeholders pushed unknown brand to about 6400 rows so I now count both. The biggest revision was the proposal itself. When we last discussed the project our instructor pointed out flaws because our main focus was the outcome and not the technical part. She appreciated our idea and gave recommendations and afterwards we used AI tools to find where we were lacking. They pointed out the technical mistakes such as the leakage risk in the description, the shifting category mix and the missing baseline and that changed the direction of our topic towards a fair comparison within each category.

What I still do not trust. I am still not sure about a decision on the chemical flag because it is a keyword search in a field that can hold several tags and about 12 percent of rows do have more than one. I would

not trust any trend after 2022 either since reporting practice clearly changed and 2026 is only a partial year. I also doubt the counterfeit field where 16666 rows say unknown which is more than half.

Limitations and next steps. This data cannot tell us whether a listing on a Pakistani marketplace is dangerous. Every row is already a dangerous product so we have no safe examples at all, and the model can only separate chemical hazards from other hazards. That is the biggest risk to our framed question, together with the fact that Pakistan appears as an origin only 130 times. Next I will build a simple category only baseline, merge the spelling variants and group rare brands then test on the latest two years and check whether the model beats the baseline.

Contribution. We cannot put a number on it because both of us worked almost equally. The technical side was mainly looked after by me while my partner did most of the testing found the loopholes and pushed us to make several changes.